23 Costcorp sells packets of mints to shop owners.

On Monday three shop owners buy mints from Costcorp.
Each shop owner buys small packets, medium packets and large packets of mints.

Alan buys 400 packets of mints.
32 % are small packets.
40 % are large packets.

Beryl buys 500 packets of mints.

$\frac{3}{10}$ are small packets.

$\frac{1}{10}$ are large packets.

Charlie buys 150 small packets of mints so that

number of small packets : number of medium packets = 3 : 4

Work out the total number of medium packets of mints these shop owners buy.
You must show all your working.

Alan: medium = 100% – 32% – 40% = 28%

	Percent	Packets	
	100	400	
÷100	1	4	÷100
×28	28	112	×28

1 mark
Alan: find 28% of 400 = 112.

Beryl: 10 – 3 – 1 = 6 tenths medium.

50	50	50	50	50	50	50	50	50	50

Each tenth = 500 ÷ 10 = 50; 6 × 50 = 300

1 mark
Beryl: 300 medium packets.

Charlie

	Small	Medium	
	3	4	
×50	150	200	×50

1 mark
150 ÷ 3 = 50.

1 mark
4 × 50 = 200.

Total = 112 + 300 + 200 = 612

1 mark
612, supported by working.

612

(Total for Question 23 is 5 marks)